Music Mental Health Analysis

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Stakeholder

The primary stakeholder for this memo is the Director of University Counseling & Wellness Services, who oversees mental-health outreach and student support strategy on campus. This stakeholder has some familiarity with data dashboards and survey-based evidence but does not work directly with statistical modeling, so they require clear interpretations rather than technical detail. They care about this analysis because understanding whether music-listening patterns are meaningfully related to depressive symptoms can help determine whether such questions should be incorporated into screening tools, campus wellness initiatives, or early-intervention outreach.

Executive Summary

This analysis examines how music-listening behaviors relate to depression symptoms using survey data from 718 respondents. We address two questions: (1) whether daily listening time varies based on depression levels and streaming platforms, and (2) whether the number of hours spent listening to music each day predicts the likelihood of reporting high depression symptoms.

For the first question, we find that individuals with higher depression scores tend to listen to more music, although this pattern differs across streaming platforms. Listening while working, insomnia symptoms, musician traits, and certain genre preferences were stronger and more consistent predictors of listening time than depression alone. For the second question, our logistic regression model shows that daily listening hours do not meaningfully increase the odds of being in a high-depression category once other factors are taken into account. Instead, anxiety and insomnia serve as the strongest predictors of elevated depression symptoms.

Overall, although depression is associated with slightly higher listening time, daily music-listening hours are not a reliable indicator of high depression risk. Co-occurring mental-health symptoms provide far more informative predictive value than listening duration.

Problem Motivation

University counseling centers today face a clear and growing challenge: many students show signs of depression or anxiety, yet only a fraction seek help. A recent large study found that roughly 48.9% of college students in the post-pandemic era report depressive tendencies (Luo et al., 2024). Another analysis of more than 100,000 students across 64 studies estimated a pooled prevalence of depression symptoms at 33.6% and anxiety symptoms at 39.0%, demonstrating how widespread mental-health concerns are in higher education (Li et al., 2022).

Because counseling centers have limited resources, and because many students may not seek formal supoort, there is an urgent need for low-burden, behavior-based signals that could help identify or screen for potential mental-health risk. One promising candidate for such indicators is everyday music-listening behavior. Music listening is a near-universal daily behavior among young adults, and research suggests it may serve as a meaningful emotional regulator or mental-health resource. Studies on music's broader health and well-being effects have documented its benefits on mood, stress, immune function, and mental-health outcomes, including among individuals with diagnosed disorders (Rebecchini et al., 2021). More recent meta-analytic reviews find that informal "musicking", not only formal therapy, consistently supports emotion regulation in non-clinical populations (Peters et al., 2024), and music-based interventions have been shown to reduce symptoms of depression and anxiety (Rodwin et al., 2022). .

Given these findings, everyday listening patterns may carry a measurable signal of underlying mental-health status, offering a low-cost way to complement traditional screening tools. If simple behaviors such as daily listening time, preferred streaming platforms, or music-use contexts reliably align with depression symptoms, counseling centers could incorporate brief questions about music habits into wellness check-ins or outreach initiatives. Such indicators would never replace clinical tools, but they could provide an accessible, early-warning layer that helps identify students who might otherwise go unnoticed.

Results and High-Level Findings

Overview of Key Findings

Across the full cleaned dataset (N = 645 Respondents for Model 1 and N = 718 Respondents for Model 2), several clear patterns emerged linking students' music-listening habits with their self-reported mental-health symptoms. While these relationships cannot establish causality, they reveal meaningful behavioral patterns that could inform low-burden wellness screening.

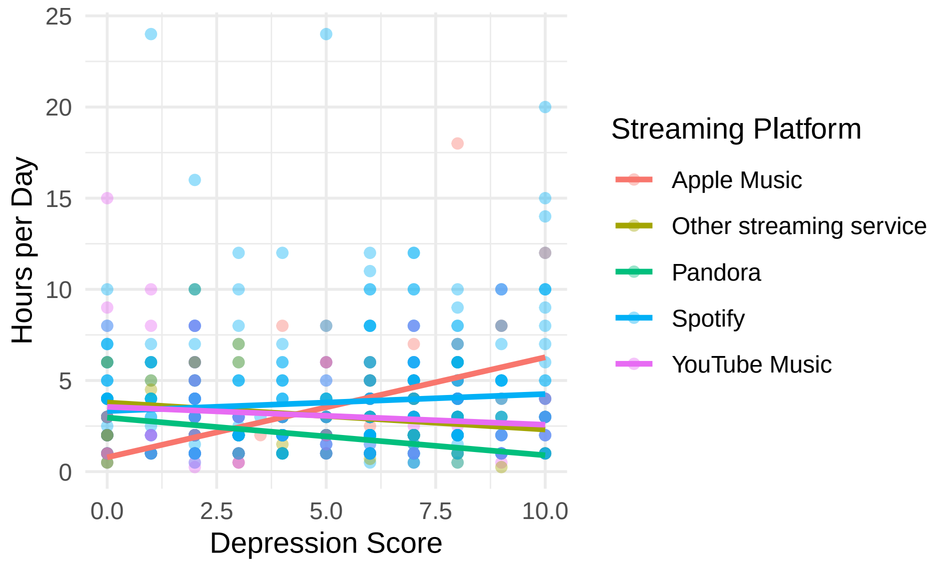

Figure 1: Daily Listening Time vs. Depression Score by Streaming Platform.

As shown in Figure 1, students with higher depression scores tend to listen to more music per day. Among Apple Music users, listening time increases from about 1 hour/day at low depression levels to 3–4 hours/day at higher levels. Spotify and YouTube Music users show smaller increases, while Pandora users display a slight decline. These differences highlight that both emotional state and platform choice shape listening behavior.

Drivers of Daily Listening Time

To understand what predicts how much students listen to music, we fit a multiple linear regression model with a log-transformed outcome: log(daily listening hours + 0.1). This transformation reduces the influence of extreme values and allows coefficients to be interpreted as percentage changes in listening time.

The final model used N = 645 respondents after filtering non-platform users and removing three influential outliers.

Term	Exp(Estimate)	Std Error	Exp(2.5% CI)	Exp(97.5% CI)	p-value
Depression	1.112	0.032	1.044	1.184	0.001
Primary.streaming.serviceOther streaming service	2.007	0.245	1.241	3.245	0.005
Primary.streaming.servicePandora	2.172	0.355	1.083	4.359	0.029
Primary.streaming.serviceSpotify	1.780	0.197	1.209	2.622	0.004
Primary.streaming.serviceYouTube Music	1.868	0.216	1.223	2.853	0.004
Age	0.995	0.003	0.990	1.000	0.043
While.workingYes	1.831	0.065	1.613	2.078	<0.001
InstrumentalistYes	0.852	0.063	0.753	0.964	0.011
ComposerYes	1.164	0.075	1.004	1.350	0.044
Insomnia	1.027	0.009	1.008	1.046	0.004
Fav.genreJazz	1.497	0.175	1.062	2.110	0.021
Fav.genreLatin	2.595	0.457	1.058	6.366	0.037
Depression:Primary.streaming.serviceOther streaming service	0.848	0.043	0.779	0.922	<0.001
Depression:Primary.streaming.servicePandora	0.813	0.067	0.713	0.926	0.002
Depression:Primary.streaming.serviceSpotify	0.902	0.033	0.845	0.963	0.002
Depression:Primary.streaming.serviceYouTube Music	0.869	0.038	0.806	0.937	<0.001

Table 1: Significant Predictors from RQ1 Linear Model (Original and Exponentiated Coefficients).

The regression results in Table 1 show several clear factors associated with higher (or lower) daily listening time after adjusting for demographic and behavioral variables. Depression remains a meaningful predictor: each one-point increase in depression score corresponds to roughly an 11 percent increase in daily listening time, indicating that students with more severe symptoms tend to engage more with music throughout the day. Platform choice is also strongly associated with listening behavior. Compared with Apple Music users, students using Spotify, YouTube Music, Pandora, or other services listen between 78 percent and 117 percent more per day, suggesting large platform-based differences in engagement.

Listening while working emerges as one of the strongest predictors in the model, with these students listening about 83 percent more per day than those who do not listen during work. Several individual characteristics also matter: composers listen about 16 percent more, whereas instrumentalists listen about 15 percent less. Insomnia shows a modest positive effect, with each point associated with a 2 to 3 percent increase in listening time. Finally, musical preferences appear relevant, as jazz listeners average roughly 50 percent more listening time and Latin listeners more than double the listening time of the reference group. Interaction terms indicate that the positive relationship between depression and listening time is weaker among students on most major platforms compared with Apple Music users. Together, these patterns highlight that emotional state, platform choice, and listening context all play meaningful roles in shaping how much music students consume daily.

Indicators of High Depression Symptoms

To assess whether music behavior can meaningfully signal elevated depression, we fit a logistic regression model predicting whether a student scored 6 or above on the depression scale using N = 718 respondents.

term	estimate	conf.low	conf.high	p.value
Hours.per.day	1.05	0.99	1.12	0.140
Anxiety	1.40	1.30	1.51	<0.001
Insomnia	1.17	1.10	1.24	<0.001

Table 2: Key Predictors from Logistic Regression (Odds Ratios)

Table 2 shows that daily listening hours are not a reliable standalone indicator of high depression. This table includes only predictors that are statistically significant (p values below 0.05), along with hours per day of listening time, since this was our primary independent variable of interest. The odds ratio for listening time is 1.05, which would mean that each additional hour of listening increases the odds of high depression by about 5 percent, but since the p-value is above our 0.05 threshold (at 0.140), we cannot say this is statistically significant.

In contrast, anxiety is the strongest predictor of high depression, with an odds ratio of 1.40 ($p < 0.001$), meaning each one-point increase in anxiety raises the odds of high depression by about 40%. Insomnia also emerges as a significant predictor, corresponding to a 17% increase in risk per hour increase in daily listening time ($p < 0.001$).

Taken together, these results show that while higher depression is associated with more listening, listening time alone should not be treated as a depression flag. Instead, anxiety and insomnia—well-validated psychological indicators—remain the most reliable behavioral cues.

Limitations and Recommended Next Steps

Although music-listening patterns carry some signal, especially the tendency for students with higher depression to listen more hours per day, the results suggest that music behavior alone is not a reliable standalone indicator of high depression risk. Any practical use of these insights should therefore be paired with validated psychological information, not used in isolation.

However, due to the nature of our data, there are two substantive limitations that are worth keeping in mind when interpreting the findings. First, all mental-health measures in the dataset rely on single self-reported items, not standardized clinical scales such as the PHQ-9 or GAD-7. These single-item scores are coarse and may under- or over-estimate true symptom severity, which limits how confidently we can infer risk or impact of certain variables from them. Second, the data are cross-sectional, meaning we observe students' music habits and mental-health symptoms at one point in time. This prevents us from determining whether increased listening is a response to worsening mood, a coping strategy, or simply a personal preference unrelated to mental-health changes.

Despite these limitations, the patterns do point toward practical next steps for wellness monitoring and outreach. Because anxiety and insomnia, which have been validated as co-indicators of depression, are far more predictive than listening time, counseling services could focus on early screening, brief questionnaires, or outreach strategies that incorporate these co-occurring symptoms.

Music-related questions may still play a useful supporting role. Although our analysis did not find strong predictive value for total listening hours alone, the limitations of the dataset mean we cannot rule out that more specific aspects of music use relate meaningfully to students' mental health. A single measure like "hours per day" does not capture when students listen, what emotional purpose the music serves, or how listening fits into their daily routines. These dimensions may still offer valuable insight.

Moving forward, the counseling center could incorporate a few brief questions into wellness screenings that ask students how they use music, such as whether they listen to manage stress, regulate mood, support sleep, or avoid intrusive thoughts. Gathering this targeted information would be low burden and could help staff identify students who may benefit from additional assessment or outreach.

The university could also consider running its own campus-wide survey that includes these improved questions and relies on more representative sampling than public social media–based datasets. Collecting higher-quality, campus-specific data would allow the counseling center to understand patterns in its own student population and evaluate whether music-related behaviors provide meaningful early signals of mental-health needs. This approach would provide a stronger evidence base for future screening tools or outreach strategies.

Data

Data Background

The data used in this analysis comes from the Music and Mental Health (MxMH) Survey Results hosted on Kaggle. The dataset was collected through an online Google Form that was distributed via social media and public forums. It contains 736 respondents (rows) and includes variables on demographic characteristics (such as age), music-listening habits (hours per day, streaming platform, frequency of listening to certain genres), and self-reported mental-health scores for anxiety, depression, insomnia, and OCD (rated 0-10).

The data is cross-sectional and self-reported, meaning it is suitable for identifying associations/correlations but not establishing causal effects.

Data Cleaning

Our data cleaning process began with an assessment of missing values across all columns. This revealed 107 null entries in the BPM column and a single null entry in the Age column. After looking further into the BPM data collection, we discovered that it was meant to represent the “Beats per minute of favorite genre.” However, rows with the same reported favorite genre displayed inconsistent BPM values, indicating that participants did not interpret or answer the question reliably. Combined with the large number of missing entries, this inconsistency made the variable unsuitable for analysis, so we dropped the BPM column entirely. The single row with a missing Age value was also removed to maintain completeness for demographic modeling.

Next, we examined the dataset for implausible or erroneous entries. During this process, we identified a respondent who reported being 89 years old and listening to music for 24 hours per day, and they were doing so while working. Because this combination is not physically possible and would exert undue influence on any model involving hours per day, we removed this outlier from the dataset.

We then reviewed all variables designed to accept only binary “Yes” or “No” responses. Several of these columns contained values outside of the allowed set, such as misspellings, alternative phrasing, or blank entries. To maintain consistency in these binary predictors, we filtered out any rows that did not provide valid “Yes” or “No” answers across all such variables, resulting in the removal of nine additional rows.

After this, we checked the music effects variable and found seven cases where respondents left this question blank. Because an empty string does not constitute a valid category and this variable is used in later modeling, we removed these seven rows as well.

For analyses involving streaming services, we examined the primary streaming service column and identified two types of unusable entries: blank responses and the explicit answer “I do not use a streaming service.” Because our first research question specifically compares listening patterns across streaming platforms, respondents without a platform could not be meaningfully included in that analysis. Therefore, for the dataset used in RQ1, we removed all rows with missing streaming-service information as well as those who reported not using any streaming service.

Finally, to prepare the binary outcome for our second research question, we created a high depression variable that flags individuals with elevated symptom levels. Because the average depression score in the dataset was 4.82 and the median was 5, we designated respondents with a score of 6 or higher as experiencing “high depression.” This threshold represents a meaningful distinction between typical responses and substantially elevated symptom reports, allowing us to model the likelihood of more severe depression symptoms.

After cleaning, there were 645 rows in the data used for model 1 and 718 rows in the data used for model 2.

Models

Research Question 1: "How does the number of hours spent listening to music per day vary by depression status and primary streaming service?"

To investigate how the number of hours spent listening to music per day varies based on depression symptoms and primary streaming service, we fit a multiple linear regression model using hours per day as the continuous response variable. Linear regression is appropriate for this question because the goal is to estimate how the average listening duration changes across different levels of depression and across streaming platforms, while adjusting for potential confounders. We also included an interaction term between depression and primary streaming service, which tests whether the association between self-reported depression symptoms and daily listening time differs by the platform a respondent uses.

Beyond the primary predictors of depression and primary streaming service, we included a set of covariates to reduce potential confounding and isolate the effects of depression and steaming service on listening time. These covariates were selected because they relate to both music-listening behavior and mental-health-related characteristics without making the model too busy. These covariates include age, while working (whether the respondent listens while working), instrumentalist (yes/no), composer (yes/no), exploratory (whether they tend to explore new music), foreign languages (whether they tend to listen to music in foreign languages), and three mental health related variables (anxiety, insomnia, OCD, all self-reported on a scale of 0-10). We also included two music-related behavioral factors, favorite genre and music effects, to account for genre preferences and how individuals perceive music impacts their emotions. Including these variables helps adjust for individual differences in music engagement, mental-health comorbidity, and daily listening contexts that could otherwise bias the estimated relationships.

Our model, which follows the linear regression framework, relies on the standard assumptions of linearity, independence of observations, homoscedasticity, and normally distributed residuals. We also assumed that there was no multicollinearity among predictors, and all of these assumptions were assessed via diagnostic plots/tests.

Research Question 2: "Does the number of hours spent listening to music each day increase the likelihood of reporting high depression symptoms?"

For our second research question, which sought to determine if the number of hours spent listening to music each day increases the likelihood of reporting high depression symptoms, we fit a logistic regression model with a binary outcome. Logistic regression is appropriate because the response variable, high depression, classifies respondents as either above or below a threshold level of self-reported depressive symptoms, and the model estimates how the log-odds of being in the high-depression category change with daily listening time.

Outside of our outcome variable of high depression and our primary independent variable, daily hours spent listening to music, we also included several covariates. Specifically, we adjusted for age, indicators of music engagement (instrumentalist and composer), co-occurring mental-health conditions (anxiety, insomnia, and OCD), and two global music-use characteristics (favorite genre and music effects). These variables may influence both how much an individual listens to music and their underlying likelihood of experiencing elevated depressive symptoms, thereby helping reduce confounding effects.

Logistic regression, and therefore our model, assumes independent observations, linearity of continuous predictors on the log-odds scale, and the absence of multicollinearity, all of which were checked via plots and tests.

Assessment and changes

Research Question 1

The first model we fit for RQ1 was a linear regression model that regressed the raw outcome of hours of music listening per day on all available predictors in the data set.

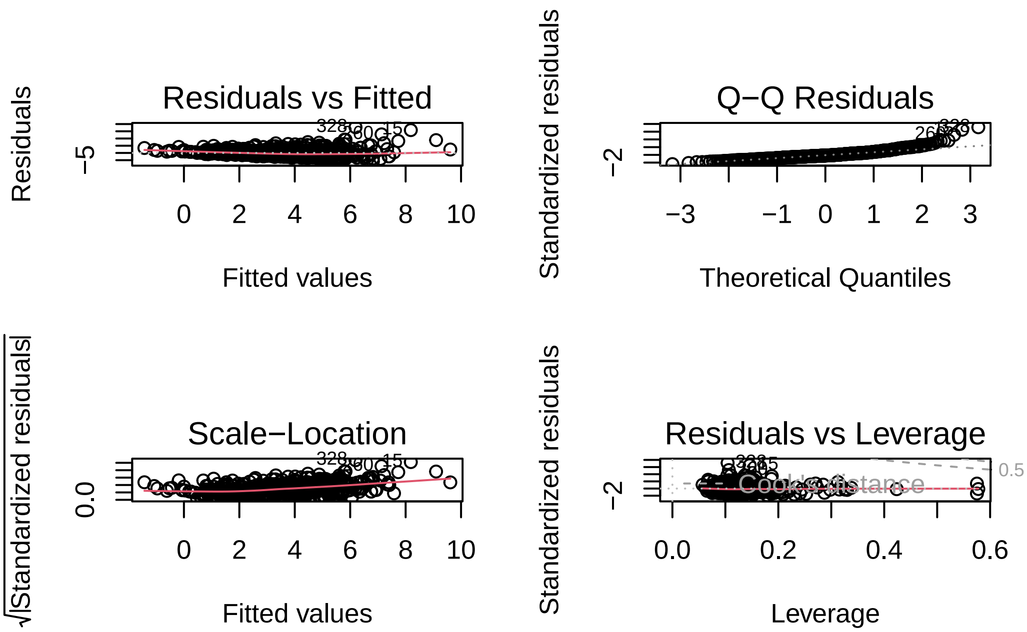

Figure 2: Diagnostic plots for the raw RQ1 model.

After fitting this initial model, we first examined the standard diagnostic plots, as shown in Figure 2: Residuals vs Fitted, Normal Q-Q, Scale-Location, and Residuals vs. Leverage. The Residuals vs Fitted plot revealed a strong violation of homoscedasticity, with the points forming a funnel-shape, and a small violation of linearity, where the red smoothing line curved upward rather than remaining flat. The Normal Q–Q plot showed noticeable departures from the 45 degree line, especially in the upper tail, a clear strong violation of normality. The Scale–Location plot further confirmed heteroscedasticity, with residual spread increasing at higher fitted values. Finally, the Residuals vs. Leverage plot identified several influential observations with large residuals and unusually high leverage, suggesting they were exerting disproportionate influence on the fitted model.

After examining the diagnostic plots, we assessed multicollinearity using GVIF values, which adjust for the degrees of freedom of multi-level categorical variables. For most predictors, including age, musician traits, mental-health indicators, and the genre-frequency variables, the adjusted $\text{GVIF}^{1/(2\cdot \text{Df})}$ values were all close to 1.1-1.3, indicating low multicollinearity and no immediate concern for inflation of variance. The interaction term between depression and primary streaming service also showed a modest adjusted GVIF value of approximately 1.10, further supporting stability in these estimates.

One exception was favorite genre, a 16-level categorical variable, which produced a very large raw GVIF (2400). However, its adjusted value, $\mathrm{GVIF}^{1/(2\cdot\mathrm{Df})} \approx 1.30$, fell well within acceptable thresholds, since high raw GVIF numbers are expected for factors with many degrees of freedom. No predictors exhibited adjusted GVIF values approaching levels (2 or 3) that would indicate problematic multicollinearity.

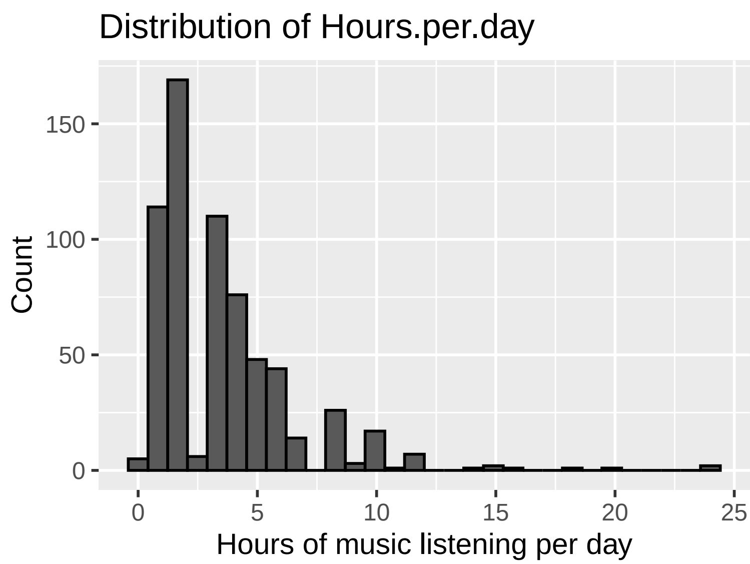

Figure 3: Diagnostic plots for the raw RQ1 model.

Because the full model included many weakly justified predictors, we fit a reduced model to improve parsimony and check whether the diagnostic issues were caused by unnecessary complexity. We also inspected a histogram of hours per day in Figure 3, which was strongly right-skewed, suggesting structural issues with the outcome itself. The reduced model produced essentially the same adjusted $R^2 (\approx 0.16)$ and similar residual behavior, with persistent violations of linearity, homoscedasticity, and normality, plus influential points. This showed that removing predictors did not improve model fit and that the underlying outcome distribution, not just model complexity, was driving the diagnostic problems.

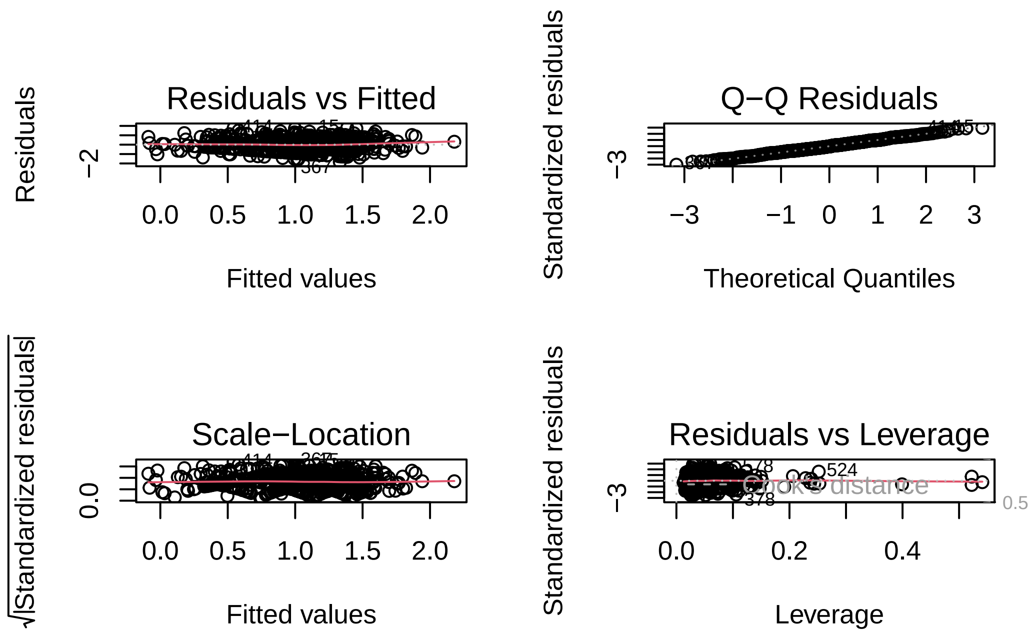

Figure 4: Diagnostic plots for the log-transformed RQ1 model.

To address this we applied a log transformation to the outcome and re–fit the model with log(Hours per day+ 0.1) as the response. On the log scale, the Residuals vs fitted plot and Q–Q plot showed substantially more symmetric and homoscedastic residuals (Figure 4) and the histogram of the transformed outcome indicated a much more balanced distribution. These improvements suggested that the log–transformed model better met the assumptions of linear regression. We also used variance inflation factors (VIFs) to check for multicollinearity among predictors and did not find strong evidence of a violation.

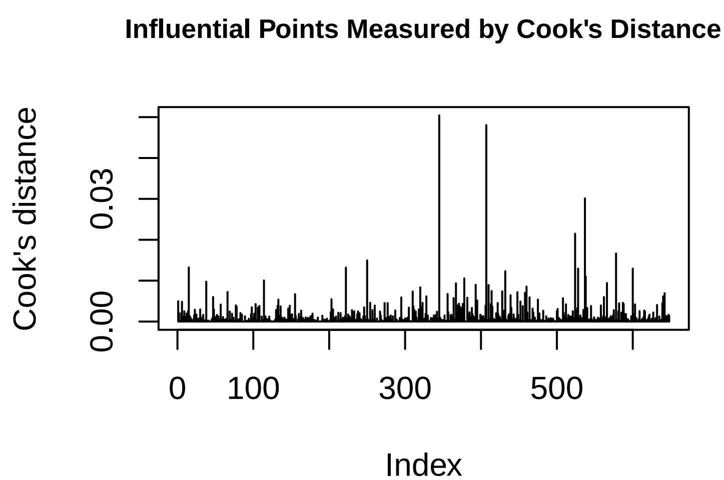

Figure 5: Influential Points Measured by Cook's Distance.

To evaluate influential observations, we computed Cook's distance for the log-transformed RQ1 model and examined the corresponding plot (Figure 5). We also inspected the residual and leverage diagnostic plots for outliers. Combining these approaches, we identified three extreme points to remove: two points were flagged as outliers in both the diagnostic plots and Cook's distance, and one additional point appeared as an outlier only in the diagnostic plots. After removing these three extreme points , we refit the log-transformed model. The residuals are now more symmetrically distributed (Min = -1.84, Max = 1.80, Median   0), and the residual standard error decreased to 0.625, indicating improved model fit.

Several predictors remain significant, including Depression, Primary streaming service, While working, Instrumentalist, Composer, Insomnia, and select favorite genres, with interactions between Depression and streaming platform also holding. The adjusted R^2 increased to 0.220, showing that the log transformation and removal of influential points slightly improved the explained variance.

Model	Adj_R2	AIC
Raw model	0.1938095	3173.908
Reduced model	0.1579996	3158.997
Log transform	0.2324734	1309.501
Log transform + remove outliers	0.2198529	1260.190

Table 3: Model comparison for RQ1 using adjusted R-squared and AIC.

Table 3 presents a comparison of the models for RQ1 using adjusted R^2 and AIC. The raw model explained about 19% of the variance but had a high AIC, reflecting poorer fit. Reducing predictors slightly lowered adjusted R^2 but improved AIC marginally. Applying the log transformation improved both adjusted R^2 (to 0.232) and AIC (to 1309), indicating better model performance. Finally, removing the three outliers slightly reduced adjusted R^2 to 0.220 but further lowered AIC to 1260, suggesting a more parsimonious and stable model. Overall, the log-transformed model with outliers removed balances interpretability, fit, and the influence of extreme points, and became our primary model for question 1.

Term	Estimate	Std Error	statistic	p-value	2.5% CI	97.5% CI
\(Intercept\)	-0.133	0.237	-0.5611308	0.575	-0.597	0.332
Depression	0.106	0.032	3.2931047	0.001	0.043	0.169
Primary.streaming.serviceOther streaming service	0.696	0.245	2.8455457	0.005	0.216	1.177
Primary.streaming.servicePandora	0.776	0.355	2.1878891	0.029	0.079	1.472
Primary.streaming.serviceSpotify	0.577	0.197	2.9249186	0.004	0.189	0.964
Primary.streaming.serviceYouTube Music	0.625	0.216	2.8988409	0.004	0.202	1.048
Age	-0.005	0.003	-2.0325505	0.043	-0.010	0.000
While.workingYes	0.605	0.065	9.3626702	<0.001	0.478	0.732
InstrumentalistYes	-0.160	0.063	-2.5549132	0.011	-0.283	-0.037
ComposerYes	0.152	0.075	2.0181945	0.044	0.004	0.300
ExploratoryYes	0.072	0.061	1.1673916	0.244	-0.049	0.192
Foreign.languagesYes	0.060	0.054	1.1261125	0.261	-0.045	0.165
Anxiety	0.000	0.011	0.0427548	0.966	-0.022	0.023
Insomnia	0.026	0.009	2.8836975	0.004	0.008	0.044
OCD	0.019	0.010	1.9530351	0.051	0.000	0.037
Fav.genreCountry	0.149	0.168	0.8892265	0.374	-0.180	0.479
Fav.genreEDM	0.248	0.149	1.6591055	0.098	-0.046	0.542
Fav.genreFolk	0.094	0.161	0.5845425	0.559	-0.223	0.411
Fav.genreGospel	0.060	0.317	0.1889684	0.850	-0.562	0.682
Fav.genreHip hop	0.134	0.153	0.8818361	0.378	-0.165	0.434
Fav.genreJazz	0.403	0.175	2.3074706	0.021	0.060	0.747
Fav.genreK pop	-0.072	0.175	-0.4120588	0.680	-0.417	0.272
Fav.genreLatin	0.954	0.457	2.0872769	0.037	0.056	1.851

Fav.genreLofi	0.060	0.227	0.2636701	0.792	-0.386	0.506
Fav.genreMetal	0.132	0.126	1.0502311	0.294	-0.115	0.378
Fav.genrePop	-0.117	0.120	-0.9754466	0.330	-0.354	0.119
Fav.genreR and B	0.128	0.153	0.8348958	0.404	-0.173	0.429
Fav.genreRap	0.283	0.176	1.6140922	0.107	-0.061	0.628
Fav.genreRock	0.097	0.115	0.8419446	0.400	-0.129	0.322
Fav.genreVideo game music	-0.266	0.156	-1.7049786	0.089	-0.573	0.040
Music.effectsNo effect	0.017	0.063	0.2704797	0.787	-0.106	0.140
Music.effectsWorsen	-0.106	0.175	-0.6037773	0.546	-0.450	0.238
Depression:Primary.streaming.serviceOther streaming service	-0.165	0.043	-3.8441528	<0.001	-0.250	-0.081
Depression:Primary.streaming.servicePandora	-0.207	0.067	-3.1180379	0.002	-0.338	-0.077
Depression:Primary.streaming.serviceSpotify	-0.103	0.033	-3.0904123	0.002	-0.169	-0.038
Depression:Primary.streaming.serviceYouTube Music	-0.140	0.038	-3.6613614	<0.001	-0.215	-0.065

Table 4: Final linear regression model for RQ1 (log-hours with influential points removed).

Table 4 shows that depression is associated with higher listening time ($\hat{\beta} = 0.106, p = 0.001$), but this effect is reduced for users of major streaming platforms, as indicated by the negative interaction terms (e.g., Spotify: $\hat{\beta} = -0.103, p = 0.002$; YouTube Music: $\hat{\beta} = -0.140, p < 0.001$). Several platforms also have positive main effects: Spotify ($\hat{\beta} = 0.577, p = 0.004$), YouTube Music ($\hat{\beta} = 0.625, p = 0.004$), Pandora ($\hat{\beta} = 0.776, p = 0.029$), and Other services ($\hat{\beta} = 0.696, p = 0.005$). All of these predict higher log-hours relative to the baseline streaming category, Apple Music.

Listening while working has the largest coefficient in the model ($\hat{\beta} = 0.605, p < 0.001$), indicating substantially higher daily listening. Additional significant predictors include insomnia ($\hat{\beta} = 0.026, p = 0.004$), being a composer ($\hat{\beta} = 0.152, p = 0.044$), being an instrumentalist ($\hat{\beta} = -0.160, p = 0.011$), age ($\hat{\beta} = -0.005, p = 0.043$), and two favorite genres: Jazz ($\hat{\beta} = 0.403, p = 0.021$) and Latin ($\hat{\beta} = 0.954, p = 0.037$). Most remaining predictors have confidence intervals crossing zero, indicating no detectable association with listening time.

Research Question 2

For RQ2, we modeled a binary outcome indicating "high depression" (depression score 6) using logistic regression, with daily listening time (Hours per day) as our primary predictor.

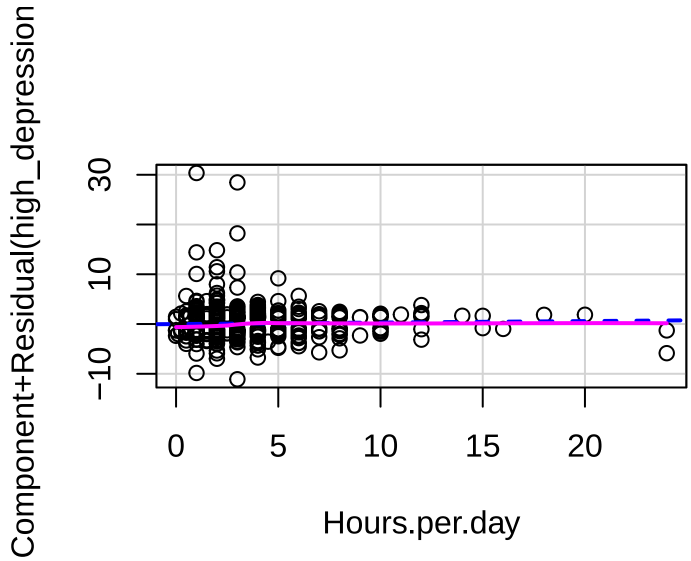

Figure 6: Partial Residuals Plot for Hours.per.day.

Because logistic regression assumes that continuous predictors have a linear relationship with the log–odds of the outcome, we began by checking this assumption for Hours per day using a component–plus–residual (partial residual) plot (Figure 6). This plot did not reveal strong curvature or systematic departures from linearity, as the pink line was fitted very tightly to the blue, so we retained Hours per day in linear form.

We then examined several additional diagnostics. First, we used VIFs to assess multicollinearity among the predictors and did not detect problematic levels of collinearity. Second, we evaluated overall model fit with a Hosmer–Lemeshow goodness–of–fit test, which revealed a p-value greater than 0.05 (0.1641), meaning the test indicates no substantial issue with the fitting of the model.

We calculated leverage values using the hat matrix and identified several observations above the common threshold of $2\bar{h}$. However, high leverage alone is not sufficient evidence to remove points in logistic regression, particularly in models with many categorical predictors. Because these cases did not show extreme Pearson residuals either, we retained all observations for the final model.

TN	FP	FN	TP	Accuracy	Kappa	Sensitivity	Specificity	Precision	Recall	F1	Balanced Accuracy
277	95	103	243	0.724	0.447	0.719	0.729	0.702	0.719	0.711	0.724

Table 5: Confusion Matrix and Classification Metrics, Threshold = 0.5.

To assess predictive performance, we computed an initial confusion matrix (Table 5) based on the conventional 0.5 probability threshold. The initial confusion matrix showed 277 true negatives, 243 true positives, 95 false negatives, and 103 false positives, yielding an overall accuracy of 0.724. Sensitivity was 0.719, meaning the model correctly identified about 72% of participants with high depression, while specificity was 0.729, indicating a similar rate of correct classification among low-depression participants. The model's precision (0.702) and F1 score (0.711) suggested moderate predictive performance but also revealed room for improvement, particularly in reducing false negatives.

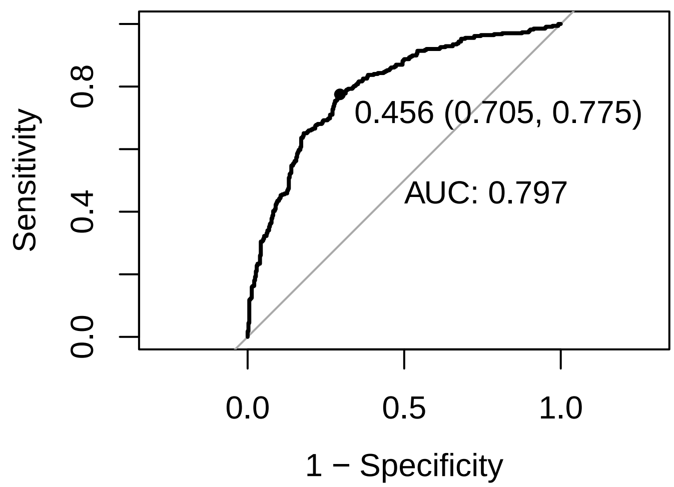

Figure 7: ROC Curve.

To further assess model performance, we examined the ROC curve (Figure 7) and computed the AUC. The model achieved an AUC of 0.797, indicating strong discriminative ability. The optimal classification threshold was identified as 0.456, which balances sensitivity and specificity more effectively than the default 0.50 cutoff.

TN	FP	FN	TP	Accuracy	Kappa	Sensitivity	Specificity	Precision	Recall	F1	Balanced Accuracy
268	76	112	262	0.738	0.478	0.775	0.705	0.701	0.775	0.736	0.74

Table 6: Confusion Matrix and Classification Metrics (Threshold = 0.456).

At this optimized threshold, we recomputed the confusion matrix (Table 6), and sensitivity increased from 0.719 to 0.775, meaning the model detected a larger proportion of high-depression cases, while specificity remained reasonably high at 0.705. Overall accuracy also improved from 0.724 to 0.738, and the F1 score increased from 0.711 to 0.736, demonstrating that threshold adjustment meaningfully enhanced the model's ability to classify high depression symptoms.

Finally, we tested whether hours per day meaningfully improved the logistic regression model by comparing the full model to a reduced model that excluded this predictor. A likelihood ratio test showed no significant improvement from adding hours per day ($\chi^2(1) = 2.23,\ p = 0.135$), indicating that daily music-listening hours do not provide additional explanatory power for predicting high depression symptoms once other factors are accounted for. Model fit indices aligned with this result: the AIC values for the reduced and full models were nearly identical (839.39 vs. 839.16), confirming that including hours per day does not meaningfully enhance model performance. However, because our research question specifically examines the role of daily listening time, we retain the full model as our primary specification.

Term	Odds Ratio	Std Error	statistic	p-value	2.5% CI	97.5% CI
\(Intercept\)	0.08	0.52	-4.8161509	p<0.001	0.03	0.22
Hours.per.day	1.05	0.03	1.4764301	0.140	0.99	1.12
Age	0.99	0.01	-1.4976696	0.134	0.97	1.00
InstrumentalistYes	0.94	0.22	-0.2893564	0.772	0.61	1.43

ComposerYes	1.13	0.26	0.4570654	0.648	0.68	1.87
Anxiety	1.40	0.04	8.6830117	p<0.001	1.30	1.51
Insomnia	1.17	0.03	5.0936882	p<0.001	1.10	1.24
OCD	0.97	0.03	-0.7990682	0.424	0.91	1.04
Fav.genreCountry	0.57	0.59	-0.9522194	0.341	0.17	1.79
Fav.genreEDM	1.18	0.53	0.3036052	0.761	0.41	3.37
Fav.genreFolk	0.79	0.53	-0.4380352	0.661	0.28	2.26
Fav.genreGospel	0.20	1.30	-1.2446057	0.213	0.01	1.99
Fav.genreHip hop	2.54	0.54	1.7321500	0.083	0.90	7.49
Fav.genreJazz	0.72	0.61	-0.5339638	0.593	0.21	2.42
Fav.genreK pop	0.41	0.61	-1.4771463	0.140	0.12	1.32
Fav.genreLatin	1.01	1.58	0.0063683	0.995	0.03	32.42
Fav.genreLofi	1.98	0.83	0.8259485	0.409	0.42	11.47
Fav.genreMetal	1.17	0.42	0.3797032	0.704	0.51	2.70
Fav.genrePop	0.66	0.41	-1.0317028	0.302	0.30	1.46
Fav.genreR and B	0.73	0.55	-0.5711555	0.568	0.24	2.13
Fav.genreRap	0.63	0.62	-0.7514045	0.452	0.18	2.09
Fav.genreRock	1.30	0.38	0.6954891	0.487	0.62	2.77
Fav.genreVideo game music	0.57	0.49	-1.1384220	0.255	0.21	1.50
Music.effectsNo effect	1.11	0.21	0.4851821	0.628	0.73	1.69
Music.effectsWorsen	2.60	0.60	1.5973620	0.110	0.84	9.14

Table 7: Logistic regression model for RQ2 (odds ratios with 95% confidence intervals).

Table 7 summarizes the corresponding regression results on the odds–ratio scale, including odds ratios, standard errors, confidence intervals, and p–values. Daily hours spent listening to music showed no significant association with high depression (OR = 1.05, 95% CI: 0.99–1.12, $p = 0.140$), indicating that each additional hour of listening time does not meaningfully change the odds of reporting elevated depression symptoms. Age was also non-significant (OR = 0.99, 95% CI: 0.97–1.00, $p = 0.134$), suggesting that depression levels did not vary systematically across ages in this sample.

The only strong and consistent predictors were mental–health variables already closely tied to depression. Anxiety had a large, significant effect (OR = 1.40, 95% CI: 1.30–1.51, $p < 0.001$), meaning that each one-unit increase in anxiety score increased the odds of high depression by approximately 40%. Insomnia also significantly increased risk (OR = 1.17, 95% CI: 1.10–1.24, $p < 0.001$). OCD symptoms were not significant (OR = 0.97, $p = 0.424$).

Neither musician status (Instrumentalist or Composer) nor any favorite–genre category significantly predicted high depression, as all confidence intervals crossed 1.0. Although "Music effects: worsen'' had a relatively large odds ratio (OR = 2.60), the estimate was imprecise (95% CI: 0.84–9.14, $p = 0.110$) and therefore not statistically reliable.

Overall, the model indicates that depression symptoms are most strongly predicted by other co-occurring mental–health indicators (anxiety and insomnia), whereas hours of music listening, demographic variables, musician traits, and musical preferences show no meaningful association with elevated depression levels.

Limitations

Our analysis has several important limitations that should be kept in mind when interpreting the results.

- **Self-reported and cross-sectional data:** All variables in the dataset, including mental-health symptoms and daily listening habits, were self-reported. This introduces potential measurement error due to inaccurate recall, inconsistent interpretation of questions, or social desirability bias. Additionally, because the data is cross-sectional, we cannot determine temporal ordering or causality. For instance, even if hours of listening were associated with depression, we cannot know whether listening more contributes to symptoms or whether individuals with higher depression simply listen to more music.
- **Limited measurement of key variables:** Several important constructs were captured using single questions (for example, music effects, listening to music while working, and favorite genre), which restricts measurement precision. Some mental-health scores may also overlap conceptually, as illnesses like anxiety and insomnia strongly correlate with depression. This makes it difficult to isolate unique events even when multicollinearity diagnostics appear acceptable.
- **Simplified representation of music behavior:** Daily hours of listening is an extremely limited measure of music engagement. It does not capture when people listen (morning vs. late night), how they listen (active vs. passive), why they listen (mood regulation, entertainment, background noise), or what they listen to within genres, all of which could meaningfully relate to mental-health outcomes. Similarly, "primary streaming service" does not capture multi-platform use or variability in content algorithms.
- **Missing potential confounders:** Several important correlated variables of depression, like socioeconomic status, employment, physical health, sleep patterns, chronic stress, and academic workload, were not measured. Their absence means our model may suffer from omitted-variable bias, limiting the interpretability of estimated relationships.

Future Work

To strengthen the results and analysis, several steps could be taken in future research:

- **Collect richer and more detailed data:** Future studies would benefit from using validated multi-item psychological scales, such as PHQ-9 for depression or GAD-7 for anxiety, rather than the single self-reported questions on a scale of 0-10. Incorporating more granular measures of music engagement, such as listening context, emotional motivation, time-of-day patterns, or track-level characteristics, would also allow for more accurate modeling of how different aspects of music behavior relate to mental-health outcomes.
- **Use longitudinal or experimental study designs:** Because the current dataset is cross-sectional, it cannot speak to temporal ordering or causal relationships. Longitudinal surveys could help determine whether music-listening habits change before shifts in mood or vice-versa. Experimental designs, such as one that involves assigning participants to specific listening conditions, would further clarify causal pathways.
- **Apply more flexible modeling approaches:** Although linear and logistic regression provided interpretable results, more flexible models that are outside the scope of this class, like generalized additive models, random forests, or gradient-boosted trees, could capture non-linear relationships and interactions that traditional parametric models may miss. These models could also improve predictive accuracy, especially if supplemented with richer features.
- **Address unmeasured confounders and broaden co-variates:** Adding variables related to sleep quality, stress, socioeconomic status, work or academic pressure, or physical health would reduce omitted-variable bias. These variables are often meaningful predictors of depression and may help clarify whether music-listening habits have independent effects.

- **Expand the scope of musical data:** Streaming-platform logs or acoustic features from tracks (tempo, energy, valence, etc.) would enable more detailed analysis of how specific music properties or listening patterns relate to mental-health symptoms. This would move the analysis beyond total hours per day and allow for deeper behavioral insights.

These limitations mean that our findings should be interpreted as descriptive patterns in this sample, rather than definitive evidence about the causal effect of music listening on depression. Nonetheless, the analysis provides a useful starting point for understanding how music–listening habits and mental–health symptoms co–vary in this population.

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